

UCE693: GROUP DESIGN PROJECT				
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Course Objective: The main objective of the project is to gain hands-on experience in tackling the planning, analysis and design issues in open-ended structural design projects while performing the analysis and design of representative structural system and components. This project work covers various aspects including planning, architectural design, geotechnical constraints, structural analysis and design and construction planning & scheduling				
During the project, various experts from industry/academics/public body are invited to deliver talks on relevant issues like: • Planning & preparation of architectural drawings of the building • Design of building frames: Load pattern, design of continuous beams/slabs and detailing of various structural components as per the relevant Indian codal guidelines • Preparation of detailed structural drawings • Design of various allied services for the building project • Preparation of general & special conditions of the contract for the project including specifications of the building based upon utility & functional aspects. • Preparation of the detailed cost estimation for the project The final project report should include the following • Description of the General Design Problem, Constraints, Functions, Design Life, and Other Relevant Considerations. • Design Assumptions, Analysis Methodologies Employed, and a Flowchart of the Design Process • Specific Design Considerations and Architectural Considerations • Design Details Including: ➤ Load Types, Loads, and Load Cases ➤ Analysis and Design Computations ➤ Deformed Shapes ➤ Member Dimensions and Reinforcement Details. • Written and Illustrated Descriptions of the Architectural, Structural Designs including the services and bill of quantities. • Conclusions				
Course Learning Objectives (CLO) Upon the completion of this course, the students will be able to: 1. Function as a member of the design team 2. Develop the general arrangement drawings 3. Produce detailed structural design & drawings and viable construction sequence 4. Produce a bill of quantities and calculate approximate construction cost				

Reference Books:

1. Varghese, P. C., Limit State Design of Reinforced Concrete, PHI Publishers (2000)
2. Jain, A.K., Reinforced Concrete-Limit State Design, Nem Chand & Bros (2014)
3. Pillai & Menon, Reinforced Concrete Design, Tata McGraw Hill Publishers(2014)

Evaluation Scheme:

Activity	Submission time line	Weightage (%)
Problem Formulation & Design Flow Chart	End of Week 4 of fifth semester	5
Midway report-1	Before the start of EST of 5 th Semester	20
Midway report-2	Before the start of MST of 6 th Semester	30
Final Assessment & Final Report & Presentation	End of 6 th Semester	45 (Report = 20 + Presentation and Viva Voce = 25)